

ECON 101—Introduction to Microeconomics
Fall 2019

Midterm II

November 11, 2019

Part I: Multiple choice

Please record your answers here.

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20

MCQ	P1	P2	Total
For instructor's use only. Do not write anything in here.			

Part I: Multiple Choice

Please record all questions in the cover sheet. Each question is worth 4 points for a total of 60 points.

Version	◇	♡	♠	♣	\$
Question					
1	(c)	(a)	(c)	(a)	(b)
2	(c)	(d)	(b)	(c)	(a)
3	(d)	(b)	(c)	(b)	(d)
4	(d)	(c)	(b)	(a)	(d)
5	(a)	(d)	(d)	(b)	(a)
6	(d)	(d)	(d)	(b)	(d)
7	(b)	(c)	(b)	(c)	(d)
8	(a)	(b)	(b)	(b)	(d)
9	(b)	(b)	(b)	(b)	(d)
10	(d)	(a)	(d)	(b)	(d)
11	(d)	(b)	(b)	(a)	(d)
12	(a)	(b)	(c)	(d)	(c)
13	(a)	(c)	(c)	(d)	(c)
14	(b)	(d)	(a)	(b)	(b)
15	(a)	(d)	(c)	(c)	(c)

Part II: Open problems

Instructions: Please do your calculations in the space provided. I will take into consideration your procedure. Please record your final answer in the space provided. There are two problems each worth 20 points.

1. (20 points) Suppose that AMD has a monopoly in the market for computer chips. AMD's total costs are given by $TC(X) = \frac{5}{2}X^2 + 15,000$. Thus, AMD's marginal cost is $MC(X) = 5X$.
 - (a) (10 points) The demand for computer chips is $X_D = 100 - 0.1P$. Find the level of output that maximizes Intel's profits. What price is Intel charging? What are its profits?

AMD is producing _____ computer chips, charging a price of _____ per chip and its profits are _____.

- (b) (10 points) What level of output would maximize total surplus in the computer chip market? Calculate the dead-weight loss generated by AMD behaving as a monopolist.

The efficient level of output is _____ and the dead-weight loss is _____.

2. (20 points) The demand for flu shots this season is given by $P = 13 - 0.0005Q$. The marginal cost of a flu shot is given by $MC = 8 + 0.0005Q$.

(a) (5 points) In a competitive market, what are the equilibrium price and quantity of flu shots?

The competitive equilibrium price is _____ and the equilibrium quantity is _____.

- (b) (10 points) Suppose that the *marginal external benefit* of flu shot is $MEB = 0.00025Q$. Compute the socially efficient quantity of flu shots.

The socially efficient quantity of flu shots is _____.

- (c) (5 points) Suppose that the government decides to give a subsidy of S per flu shot. What subsidy S should the government set to achieve the social optimum?

The per-flu shot subsidy that the government needs to put is _____.